1) Soap prime at server side:

```
package server;


import javax.jws.WebService;
import javax.jws.WebMethod;
import javax.jws.WebParam;


@WebService(serviceName = "Primeservice")
public class Primeservice {
   @WebMethod(operationName = "isprime")
   public boolean isprime(@WebParam(name = "number") int number) {
      if(number<=1)
         return false;
      for(int i=2;i<=Math.sqrt(number);i++)
      {
         return false;
   }
      return true;
   }
}
```

Soap prime number at client side:

```
package ty;
import java.util.*;
public class prime {
   public static void main(String[] args)
   {
      client.Primeservice_Service service=new client.Primeservice_Service();
```

```
    client.Primeservice prime=service.getPrimeservicePort();
    Scanner sc=new Scanner(System.in);
    System.out.print("enter the number");
    int number=sc.nextInt();

    boolean result=prime.isprime( number);
    if(result)
       System.out.println(number+"is a prime number");
       else
       System.out.println(number+"is not prime number");
    }

}
```

2) <u>Soap Multiplication of two numbers at server side:</u>

```
package server;

import javax.jws.WebService;
import javax.jws.WebMethod;
import javax.jws.WebParam;

@WebService(serviceName = "multiplication")
public class multiplication {
   @WebMethod(operationName = "multiply")
   public int multiply(@WebParam(name = "a") int a, @WebParam(name = "b") int b)
{
        return a*b;
   }
}
```

Soap Multiplication of two numbers at client side:

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
  <body>
    <h2> Multiply two numbers </h2>
    <form method="post">
      number 1:<input type="text" name="n1"><br><br>
      number 2:<input type="text" name="n2"><br><br>
      <input type="submit" value="Multiply">
    </form>
    <%
      if(request.getParameter("n1")!=null)
      {
        int a=Integer.parseInt(request.getParameter("n1"));
        int b=Integer.parseInt(request.getParameter("n2"));
        client.Multiplication_Service service=new client.Multiplication_Service();
        client.Multiplication port=service.getMultiplicationPort();
        int result=port.multiply(a, b);
        out.println("<h3>Result="+result+"</h3>");
      }
      %>

  </body>
</html>
```

3) Soap addition of three numbers at server side:

```
package server;

import javax.jws.WebService;
import javax.jws.WebMethod;
import javax.jws.WebParam;

@WebService(serviceName = "AddService")
public class AddService {

   @WebMethod(operationName = "addthree")
   public int addthree(@WebParam(name = "a") int a, @WebParam(name = "b") int
b, @WebParam(name = "c") int c) {

      return a+b+c;
   }
}
```

Soap addition of three numbers at client side:

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>

   <body>
      <h2> Addition of three numbers </h2>
      <form method="post">
         Number 1:<input type="text" name="n1"><br>
         Number 2:<input type="text" name="n2"><br>
         Number 3:<input type="text" name="n3"><br>
         <input type="submit" value="add">
      </form>
      <%
         if(request.getParameter("n1")!=null)
         {
            int a=Integer.parseInt(request.getParameter("n1"));
            int b=Integer.parseInt(request.getParameter("n2"));
            int c=Integer.parseInt(request.getParameter("n3"));
```

```
            client.AddService_Service service=new client.AddService_Service();
            client.AddService port=service.getAddServicePort();
            int result=port.addthree(a, b, c);
            out.println("<h3>Result="+result+"</h3>");

            }
        %>
    </body>
</html>
```

4) Soap calculator addition and subtraction at serverside:

```
package server;

import javax.jws.WebService;
import javax.jws.WebMethod;
import javax.jws.WebParam;


@WebService(serviceName = "calculatorservice")
public class calculatorservice {


    @WebMethod(operationName = "add")
    public int add(@WebParam(name = "a") int a, @WebParam(name = "b") int b) {

        return a+b;
    }


    @WebMethod(operationName = "subtract")
    public int subtract(@WebParam(name = "a") int a, @WebParam(name = "b") int b)
{

        return a-b;
    }


}
```

Soap calculator addition and subtraction at client side:

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>

   <body>
      <h2>soap calculator </h2>
      <form method="post">
         Number 1:<input type="text" name="n1"><br>
          Number 2:<input type="text" name="n2"><br>
          <input type="submit" name="op" value="ADD">

          <input type="submit" name="op" value="subtract">
      </form>
      <%
         if(request.getParameter("op")!=null)
         {
            int a=Integer.parseInt(request.getParameter("n1"));
            int b=Integer.parseInt(request.getParameter("n2"));
            String op=request.getParameter("op");
            client.Calculatorservice_Service service=new
client.Calculatorservice_Service();
            client.Calculatorservice port=service.getCalculatorservicePort();
            int result=0;
            if(op.equals("ADD"))
               result=port.add(a, b);
            else
               result=port.subtract(a, b);
            out.println("<h3>Result="+result+"</h3>");

         }
         %>
   </body>
</html>
```